SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Case Study

Code of Conduct:

I Moshitha Kolanjiyappan (192421205) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Moshitha K

Annexure A

Short Answer Test

1, A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

A hospital can use a Naïve Bayes classifier to calculate the probability of a disease based on symptoms and medical history.

Bayes Theorem:
Probability of Disease given Symptoms = Probability of Symptoms given Disease × Probability of Disease / Probability of Symptoms.

A Bayesian Belief Network can represent relationships between disease, symptoms, age, and medical history. It can handle uncertainty by assigning probabilities instead of using fixed rules. Conditional dependencies show how one symptom or medical factor can influence another. Probability learning estimates these probabilities from historical patient data. The system can be evaluated using accuracy, precision, recall, and F1-score. Real-world challenges include missing data, noisy medical records, privacy, and the need for medical validation

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

A spam detection system can use the Naïve Bayes classifier to calculate the probability that an email is spam based on words and other features.

The system calculates the probability of Spam given the observed words. It then compares the probability of spam with the probability of a legitimate email and selects the class with higher probability.

The Bayes Optimal Classifier considers multiple possible hypotheses and selects the classification with the lowest expected error.

Maximum Likelihood Estimation estimates feature probabilities directly from training data. Minimum Description Length selects a simpler model by minimizing the total description length of the model and data. MLE may use many features and can overfit, while MDL prefers simpler models. Larger sample sizes generally provide more reliable probability estimates. Accuracy, precision, recall, and F1-score can be used to evaluate the system.

3. A retail company wants to segment its customers based on purchasing behavior, but the data contains hidden patterns and incomplete information. Apply the EM algorithm to identify clusters and estimate underlying distributions. Analyze convergence behavior, initialization challenges, and the role of probability learning. Discuss how the results can be used for targeted marketing and evaluate limitations of the approach.

The EM algorithm can be used to divide customers into groups based on purchasing frequency, spending, and product preferences.

In the Expectation step, the algorithm estimates the probability that each customer belongs to each cluster.

In the Maximization step, it updates the cluster parameters using these probabilities.

These two steps are repeated until the model converges.

EM can identify hidden customer groups even when the group labels are unknown. The initial values are important because poor initialization can lead to a local optimum. Probability learning estimates the characteristics of each customer group. The results can be used for targeted advertisements, personalized offers, and customer analysis. Limitations include sensitivity to initialization, incomplete data, noisy data, and convergence to a local optimum.

4. A research team is developing a concept learning system to classify medical images into normal and abnormal categories. Using hypothesis space search and the Candidate Elimination approach, design a model that learns from labeled examples. Analyze the impact of finite vs infinite hypothesis spaces, sample complexity, and mistake bound model on learning performance. Evaluate how well the system generalizes to unseen data.

A concept learning system can classify medical images as normal or abnormal using labeled training examples.

The Candidate Elimination algorithm maintains two boundaries:

S represents the most specific hypotheses.

G represents the most general hypotheses.

Positive examples make the specific hypothesis more general, while negative examples make the general hypothesis more specific. The remaining possible hypotheses form the version space.

A finite hypothesis space makes learning easier because the number of possible hypotheses is limited. An infinite or very large hypothesis space requires more training data and computation. Sample complexity represents the amount of data required to learn a reliable hypothesis. The mistake bound represents the possible number of errors made before learning the correct concept. Good training data helps the model generalize to unseen medical images.

5. A financial institution needs to make investment decisions under uncertain market conditions. Using Bayesian inference, Gibbs algorithm, and probability learning, design a system to predict outcomes and update beliefs dynamically. Analyze how prior knowledge influences predictions, evaluate risk using posterior probabilities, and discuss how the model adapts as new data becomes available. Consider computational challenges and optimization strategies.

A financial prediction system can use Bayesian inference to combine historical market data with prior knowledge.

Bayes Theorem is used to calculate the probability of a particular market outcome based on the available evidence.

The Gibbs algorithm can be used to estimate complex probability distributions by repeatedly sampling each variable based on the values of other variables.

Prior knowledge can strongly influence predictions when only limited data is available. Posterior probabilities can be used to estimate investment risk. When new market data becomes available, the system updates its beliefs and produces new predictions.

The main challenges are large datasets, high computational cost, and slow convergence. Optimization techniques, efficient sampling, and parallel processing can improve performance.

6. An autonomous vehicle system must make real-time driving decisions under uncertain environmental conditions such as weather, traffic, and pedestrian movement. Using Bayesian inference, Naïve Bayes classifiers, and probabilistic reasoning, design a model to predict safe driving actions. Analyze how uncertainty and sensor noise are handled, and evaluate system reliability under dynamic conditions.

An autonomous vehicle can use Bayesian inference and probabilistic reasoning to make safe driving decisions under uncertain conditions.

The system can analyze weather, traffic, pedestrian movement, road conditions, and sensor information. Naïve Bayes can classify situations as safe, caution, or dangerous.

Bayesian reasoning allows the system to update its beliefs whenever new camera, radar, or lidar information is received. Sensor noise is handled by assigning probabilities to observations rather than assuming that every sensor reading is completely accurate.

System reliability can be evaluated using accuracy, false-negative rate, response time, and safety testing. Continuous sensor updates and sensor fusion are important for handling dynamic environments.

7. A healthcare organization aims to predict patient risk levels for chronic diseases using historical and lifestyle data. Apply probability learning, Bayes' Theorem, and hypothesis space search to design the model. Analyze the impact of prior probabilities, feature dependencies, and sample size. Evaluate the model's interpretability and its usefulness in preventive healthcare strategies.

A healthcare system can use Bayes Theorem to predict the probability of developing a chronic disease based on age, lifestyle, family history, and other health-related factors.

The prior probability represents the initial probability of a disease. The evidence from patient features is then used to calculate the updated probability.

Probability learning estimates model parameters from historical patient data. Hypothesis space search can be used to compare different possible predictive models and select a suitable hypothesis.

Feature dependencies are important because factors such as obesity, blood pressure, and lifestyle may be related. Larger sample sizes usually provide more reliable probability estimates. The model can be useful for preventive healthcare because doctors can identify high-risk patients and provide early interventions.

8. A bank wants to detect fraudulent transactions using machine learning under uncertain and evolving patterns. Using Bayesian Belief Networks and EM algorithm concepts, design a detection system. Analyze hidden variables, incomplete data handling, and convergence issues. Evaluate how probability-based reasoning improves detection accuracy and reduces false positives in real-world scenarios.

A bank can use a Bayesian Belief Network to detect fraudulent transactions. The network can include transaction amount, location, device, transaction time, customer behavior, and fraud status.

Hidden variables can represent unknown customer or transaction behavior. The EM algorithm can estimate these hidden variables when some information is missing.

In the Expectation step, hidden values are estimated. In the Maximization step, model parameters are updated. These steps continue until convergence.

EM may converge to a local optimum, so different initial values can be tested. Bayesian reasoning provides a probability of fraud rather than simply giving a yes or no result. This helps reduce false positives and improves fraud detection accuracy.

9. An e-commerce platform aims to recommend products to users based on browsing and purchase history. Using Naïve Bayes and Bayesian learning concepts, design a recommendation model. Analyze how conditional probabilities influence recommendations and how uncertainty in user preferences is managed. Evaluate scalability and performance in handling large, dynamic datasets.

An e-commerce platform can use Naïve Bayes and Bayesian learning to recommend products based on browsing history, purchase history, ratings, and product categories.

The system calculates the probability that a user will prefer a particular product based on previous user behavior.

Conditional probabilities show how different behaviors influence the recommendation. For example, repeatedly viewing a particular category can increase the probability that the user will purchase products from that category.

Bayesian learning can update recommendations when new browsing or purchasing information becomes available. The system can rank products according to their probability of being preferred. For large datasets, efficient data processing and scalable computing methods are required. Cold-start users and changing user preferences are important challenges.

10. A research institute is building a system to predict climate changes using historical environmental data. Apply probabilistic models such as Bayesian inference and EM algorithm to handle incomplete and noisy datasets. Analyze convergence behavior, uncertainty modeling, and the influence of prior knowledge. Evaluate the system's effectiveness in long-term environmental forecasting.

A climate prediction system can use Bayesian inference to analyze historical temperature, rainfall, atmospheric pressure, greenhouse gas levels, and other environmental data.

Bayesian inference combines prior climate knowledge with observed data to produce updated predictions.

The EM algorithm can be used when environmental datasets contain missing values or hidden variables. In the Expectation step, missing or hidden information is estimated. In the Maximization step, model parameters are updated.

The algorithm continues until convergence. Bayesian models represent uncertainty using probability distributions. Noisy and incomplete environmental data can reduce prediction accuracy. The system can be evaluated using Mean Absolute Error, Root Mean Square Error, and prediction intervals.

11. A telecommunications company aims to predict customer churn using historical usage data and service feedback. Using probabilistic models such as **Naïve Bayes**, **Bayesian inference**, and hypothesis space search, design a predictive system. Analyze how prior probabilities and feature dependencies influence churn prediction. Evaluate how uncertainty in customer behavior is handled and discuss the effectiveness of the model in improving customer retention strategies.

A telecommunications company can use Naïve Bayes to predict whether a customer is likely to leave the service.

The model can use features such as call usage, internet usage, complaints, service quality, contract type, and customer feedback.

Bayesian inference combines the prior probability of customer churn with the observed customer information. Feature dependencies must be considered because variables such as poor service quality and customer complaints may be related.

Hypothesis space search can compare different predictive hypotheses and select the best model. Probability allows the system to express the likelihood of churn rather than giving only a yes or no result. High-risk customers can then receive special offers, improved plans, or customer support to improve retention.

12. A smart healthcare system is designed to monitor patients in real-time using wearable sensors. Using **Bayesian Belief Networks** and probability learning, develop a model to detect anomalies in vital signs. Analyze how conditional dependencies between physiological parameters are modeled and how uncertainty due to sensor noise is handled. Evaluate system performance and discuss challenges in real-time deployment.

A smart healthcare system can use a Bayesian Belief Network to monitor vital signs such as heart rate, blood pressure, oxygen level, and body temperature.

The network represents relationships between physiological parameters. For example, changes in blood pressure may influence the probability of certain heart-related conditions.

Probability learning can estimate these relationships using historical patient data. Sensor noise can be handled by assigning confidence or probability to each measurement.

The system continuously updates its prediction when new sensor readings are received. Performance can be evaluated using sensitivity, specificity, precision, recall, and response time.

Challenges include missing sensor readings, noisy data, connectivity problems, computational limitations, and false alarms.

13. A logistics company wants to optimize delivery routes under uncertain traffic and weather conditions. Using **Bayesian inference** and probabilistic reasoning, design a

system that predicts optimal routes dynamically. Analyze how prior knowledge and real-time data updates influence decision-making. Evaluate computational complexity and discuss how uncertainty impacts route optimization efficiency.

A logistics company can use Bayesian inference to predict travel time and traffic conditions.

The system can use historical traffic data along with real-time information such as traffic density, accidents, weather, and road conditions.

Prior knowledge provides an initial estimate of traffic conditions. When new real-time information becomes available, Bayesian inference updates the probability of different traffic conditions.

The system can then select the route with the lowest expected travel time or cost. Uncertainty can cause route conditions to change quickly, so continuous updates are required. Large road networks also increase computational complexity. Shortest-path algorithms, heuristic methods, and efficient data processing can improve route optimization.

14. A cybersecurity system is developed to detect network intrusions using probabilistic learning methods. Apply **Naïve Bayes classifiers**, **EM algorithm**, and Bayesian learning to model normal and abnormal network behavior. Analyze how hidden patterns and incomplete data are handled. Evaluate the system's effectiveness in reducing false alarms and improving detection accuracy in dynamic environments.

A cybersecurity system can use Naïve Bayes to classify network activity as normal or abnormal.

The system can analyze features such as packet size, connection duration, protocol type, traffic frequency, and source address.

The EM algorithm can identify hidden patterns when some network data is incomplete or unlabeled. Bayesian learning can update the model when new types of attacks are detected.

The EM algorithm may have convergence problems and can reach a local optimum. Different initializations can help improve the result.

The system can provide a probability score for each network activity. A suitable threshold can be selected to reduce false alarms while maintaining high intrusion detection accuracy. Accuracy, precision, recall, F1-score, and false-positive rate can be used for evaluation.

15. An agriculture-based system aims to predict crop yield under uncertain environmental conditions such as rainfall, soil quality, and temperature. Using **Bayesian inference** and probability learning, design a predictive model. Analyze how prior agricultural knowledge and seasonal variations influence predictions. Evaluate the model's reliability and discuss its impact on decision-making for farmers and policy planning.

An agricultural system can use Bayesian inference to predict crop yield using rainfall, temperature, soil quality, irrigation, fertilizer usage, and historical crop yield.

Prior agricultural knowledge provides an initial expectation about crop yield. New weather and soil information can then update the prediction.

Probability learning can identify relationships between environmental conditions and crop yield. Seasonal variations can be included in the model by considering different seasons and weather patterns.

The Bayesian approach is useful because it provides both the expected crop yield and the uncertainty of the prediction. Farmers can use the results to make decisions about irrigation, fertilizers, and crop selection. Policymakers can use the predictions for food supply planning. Limitations include missing weather data, changing climate conditions, and uncertain environmental measurements.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand